

User Manual for DSP Based Physiological Stress Monitoring System

Project Title:

**A Digital Signal Processing Approach to Physiological Stress Monitoring
via Skin Conductance Analysis**

Prepared for
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1 Introduction

This user manual provides instructions for operating the DSP-based physiological stress monitoring system developed using a GSR (Galvanic Skin Response) sensor and Arduino UNO.

The system measures changes in skin conductance caused by physiological responses and processes the acquired signal using a Gaussian FIR Low Pass Filter. The filtered signal is analyzed with respect to a calibrated baseline value to indicate changes in stress response.

2 System Overview

The system consists of the following modules:

- GSR Sensor for measuring skin conductance.
- Arduino UNO for signal acquisition and processing.
- Gaussian FIR Low Pass Filter for noise reduction.
- LED indicators for visual feedback.
- Buzzer for audio alert.

The basic operation flow is:

GSR Sensor → Arduino ADC → FIR Filter → Stress Analysis → LED/Buzzer Output

3 Hardware Requirements

Component	Quantity
Arduino UNO	1
GSR Sensor	1
Green LED	1
Yellow LED	1
Red LED	1
Buzzer	1
Resistors	Required
USB Cable	1

Table 1: Required Hardware Components

4 Hardware Connection

The connections between Arduino UNO and external components are:

Component	Arduino Pin
GSR Sensor Output	A0
Green LED	D2
Yellow LED	D3
Red LED	D4
Buzzer	D7

Table 2: Pin Configuration

5 Software Requirements

The following software tools are required:

- Arduino IDE
- Python with NumPy and SciPy libraries
- Serial Monitor / Serial Plotter

6 Installation and Setup

Follow these steps to prepare the system:

1. Install Arduino IDE on the computer.
2. Connect Arduino UNO to the computer using a USB cable.
3. Open the Arduino project file.
4. Select the correct board:
 - Board: Arduino UNO
 - Port: Connected Arduino port
5. Upload the program to Arduino UNO.
6. Open Serial Monitor or Serial Plotter.

7 Operating Procedure

Follow the steps below to operate the system:

1. Power on the Arduino UNO.
2. Place fingers properly on the GSR sensor electrodes.
3. Keep the hand steady during calibration.
4. Wait until the system completes baseline calibration.
5. After calibration, the system starts monitoring automatically.
6. Observe LED and buzzer responses according to signal variation.
7. Use Serial Plotter to observe raw and filtered GSR signals.

8 Calibration Procedure

Calibration is required because skin conductivity varies from person to person.

During calibration:

- Keep fingers placed on the sensor.
- Avoid unnecessary movement.
- The system collects initial samples.
- The average value is stored as the baseline.

The stress variation is calculated as:

$$\text{Stress Level} = |\text{Filtered Signal} - \text{Baseline}|$$

9 Output Interpretation

The system uses three LEDs and a buzzer to indicate different conditions.

Output	Condition
Green LED ON	Normal response
Yellow LED ON	Moderate variation
Red LED ON + Buzzer	High variation

Table 3: Output Meaning

10 Serial Plotter Monitoring

The Serial Plotter displays the processed GSR signal.

The plotted signals include:

- Raw GSR signal
- Filtered GSR signal
- Baseline value

The filtered signal should appear smoother than the raw signal due to the FIR filtering operation.

11 Troubleshooting

Problem	Possible Solution
No Serial output	Check USB connection and correct COM port.
Unstable readings	Keep fingers steady and clean the sensor surface.
LED not working	Check wiring and resistor connection.
Incorrect stress indication	Repeat calibration procedure.

Table 4: Troubleshooting Guide

12 Limitations

- The system measures physiological response only.
- It does not directly determine emotions or detect lies.
- Results may vary between different individuals.
- External noise may affect sensor readings.

13 Safety Instructions

- Do not apply excessive force on the sensor electrodes.
- Keep the circuit away from water.
- Disconnect power before changing connections.
- Handle Arduino and electronic components carefully.

14 Conclusion

This user manual describes the operation and maintenance procedure of the DSP-based GSR stress monitoring system. Following the instructions provided in this manual, users can properly set up, calibrate, and operate the system for real-time physiological signal monitoring.